Network Performance Analysis — Lab Sheet 1

Neil Lawrence

Lab

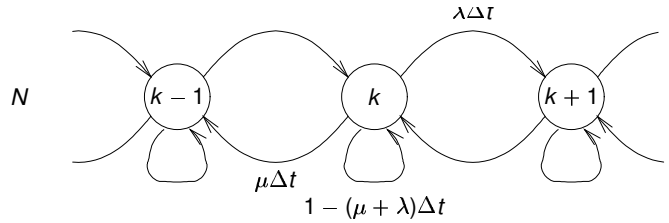
All lab classes in this course are

In this Lab you will build a discrete time simulation of a queue in MATLAB. You will investigate the statistics associated with this simulation and compare them with analytically computed values.

1. Your simulation will consist of a discrete Markov model based on a birth death process.

   ```
   % pseudo code
   deltat = 0.01;
   ```

   Use discrete time-steps of $\Delta t = 0.01$.

   .

   Code a simulation of the queue with a birth rate of $\lambda = 1$ for all states $k = 0, \ldots, \infty$, and a death rate of $\mu = 2$ for the states[1] $k = 1, \ldots, \infty$, *i.e.* at each time step, the probability of changing state by $+1$ is equal to $\lambda\Delta t$, the probability of changing state by -1 is equal to $\mu\Delta t$ and the probability of no change of state is $1 - (\lambda + \mu)\Delta t$. Run your queueing model for 1000 seconds (equivalent to $100,000$ iterations or time steps), store the state of the queue at each time step. This stored set of states can be viewed as a set of samples from the probabilistic model that your queue represents.

   [*While you wait for simulations to run you might like to start looking at the questions below*].

2. Produce a normalized histogram of the samples.

   [*Hint: A normalized histogram is one where the area of the bars adds up to one instead of the number of counts.*]

[1] $\mu_0 = 0$ to prevent negative numbers in the queue.

3. Compute the mean of the samples (the sample mean).

4. Use the detailed balance equations to derive the steady-state probabilities for each state in your queueing system. Comment on the similarities between the equations and those of the $M/M/1$ queue.

5. Use these steady-state probabilities to derive the distribution's mean.

6. Compare the computed probabilities from 4 to the normalized frequency counts from the histogram found in 2 by plotting them together. Compare the mean of your samples from 3 to the analytical mean computed in 5. Comment on differences between the analytically computed values and those computed from the samples.

7. Run your model again with $\lambda = 1.9$, keeping $\mu = 2$. Repeat steps 2 to 3 for these new values. Are the sample based estimates more or less accurate for this system? Comment on why they are more or less accurate.

8. In this lab sheet you have developed a simulation of a simple Markov birth death process. You have examined the statistics of your simulation and compared them to analytically computed values.

   Real networks are more complex than the simple network we describe here. Many cannot be studied analytically and therefore their statistics must be determined through simulation. Summarize what you have learned from the simple example above about the possible dangers of such simulations.

Useful matlab commands.

```
hist
bar
rand
```